Source photographs and capitulum localisation

Cirsium pannonicum | *Cirsium mohavense* | *Cirsium subcoriaceum*



Stored detector boxes identify candidate heads; photographs illustrate the measurement pipeline.

Continuous measurements from the boxed head in (a)

Orientation

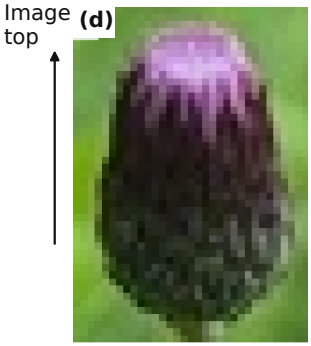


Image angle	0.7°

0° top · 90° horizontal
180° bottom

Colour



Lightness, L*	56.5
Chroma, C*	33.3
Circular hue	300.2°

Outline



Aspect ratio	1.59
Circularity	0.77
Solidity	0.95

Width-profile CV: Table S2.1

Eight constructs; hue is represented by two circular coordinates. Definitions: Table S2.1.

Eligible head measurements → photograph-level medians by domain

Source links, taxon, location and joint coordinates are retained.

Among-taxon climate gradients	Within-range colour gradients	Climate-analogue joint resemblance

Photo sources: Bobo-X (a, d–f; CC BY 4.0); Jesse Rorabaugh (b; CC0 1.0);
Diego Blanco (c; CC0 1.0). Cropped and annotated; source links are in the caption.